Planar and Dual Graphs

Planarity

The main theme of this Chapter is to answer following important question:

Is it possible to draw G in a plane without its edges crossing over?

This question of planarity is significant both from theoretical and practical perspective.

Remember the **3 utilities puzzle** from Chapter 1, the solution to which depends on finding whether the graph can be drawn on a plane.

COMBINATORIAL VERSUS GEOMETRIC GRAPHS

There are two ways of defining graphs:

One is the abstract way, which is known as the **Combinatorial definition** of the graph. And the other is the pictorial way, also known as the **Geometric definition** of the graph.

For example, an abstract graph G_1 can be defined as

$$G_1 = (V, E, \Psi)$$

where the set V consists of the five objects named a, b, c, d, and e, that is,

$$V = \{a, b, c, d, e\}$$

COMBINATORIAL VERSUS GEOMETRIC GRAPHS

And the set E consists of seven objects (none of which is in set V) named 1, 2, 3, 4, 5, 6, and 7, that is,

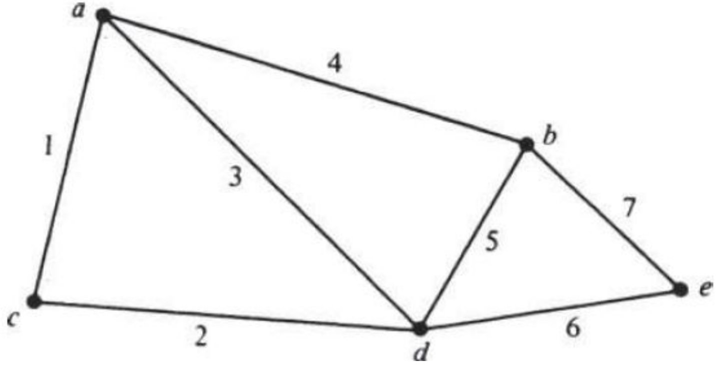
$$E = \{1, 2, 3, 4, 5, 6, 7\}$$

And the relationship between the two sets is defined by the mapping Ψ, which is

$$\Psi = \begin{bmatrix} 1 \longrightarrow (a, c) \\ 2 \longrightarrow (c, d) \\ 3 \longrightarrow (a, d) \\ 4 \longrightarrow (a, b). \\ 5 \longrightarrow (b, d) \\ 6 \longrightarrow (d, e) \\ 7 \longrightarrow (b, e) \end{bmatrix}$$

COMBINATORIAL VERSUS GEOMETRIC GRAPHS

This combinatorial abstract object G_1 can also be represented by means of a geometric figure, such as the one given below



But this merely one (out of infinitely many) representation of the graph G_1 and not the graph G_1 itself.

We could have, for instance, twisted some of the edges or could have placed e within the triangle a, d, b and thereby obtained a different figure representing G1.

PLANAR GRAPHS

A graph G is said to be planar if there exists some geometric representation of G which can be drawn on a plane such that no two of its edges intersect.

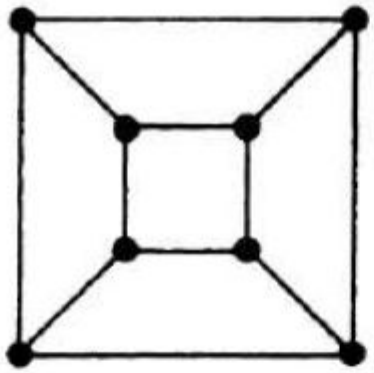
A drawing of a geometric representation of a graph on any surface such that no edges intersect is called **embedding**.

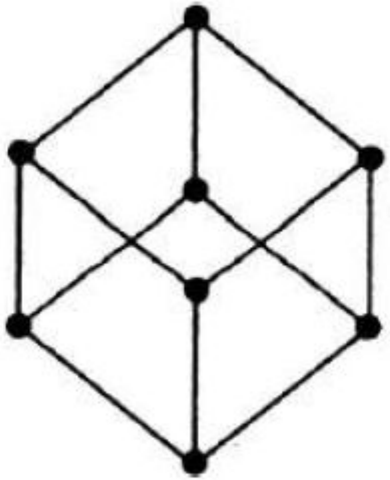
Thus, to declare that a graph G is nonplanar, we have to show that of all possible geometric representations of G none can be embedded in a plane.

PLANAR GRAPHS

These two isomorphic diagrams are different geometric representations of the same graph.

One of the diagrams is a plane representation; the other one is not. But the graph, of course, is planar.

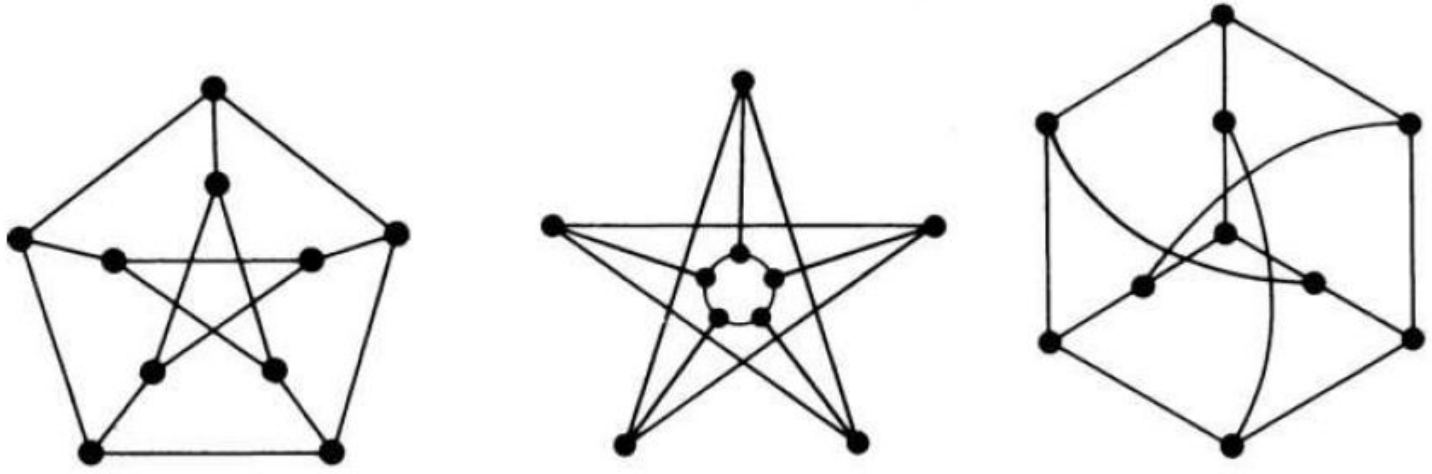




PLANAR GRAPHS

On the other hand, you will not be able to draw any of the three configurations of the graph below on a plane without edges intersecting.

The reason is that the graph which these three different diagrams represent is nonplanar.



KURATOWSKI'S TWO GRAPHS

THEOREM 5-1: "*The complete graph of five vertices is nonplanar.*"

Let the 5 vertices in the complete graph be named v_1, v_2, v_3, v_4, and v_5.

A complete graph is a simple graph in which every vertex is joined to every other vertex by means of an edge, which means there must be a circuit going from v_1 to v_2 to v_3 to v_4 to v_5 to v_1 resulting in a pentagon.

This pentagon must divide the plane of the paper into two regions, one inside and the other outside, say R_1 and R_2 respectively.

KURATOWSKI'S TWO GRAPHS

Since vertex v_1 is to be connected to v_3 by means of an edge, this edge may be drawn **inside** or **outside** the pentagon (and **both are valid**, as none will cause any intersection as of yet). Suppose that we choose to draw a line from v_1 to v_3 **inside** the pentagon.

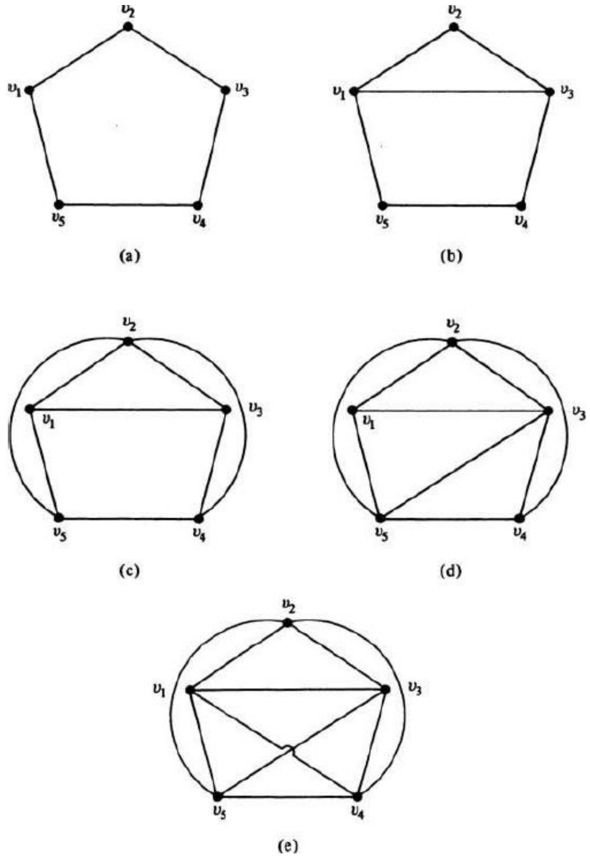
Now we have to draw an edge from v_2 to v_4 and another one from v_2 to v_5. Since **neither of these edges can be drawn inside** the pentagon without crossing over the edge already drawn, we draw both these edges **outside** the pentagon.

The edge connecting v_3 and v_5 **cannot be drawn outside** the pentagon without crossing the edge between v_2 and v_4. Therefore, v_3 and v_5 have to be connected with an edge **inside** the pentagon.

KURATOWSKI'S TWO GRAPHS

Now we have yet to draw an edge between v_1 and v_4. This edge **cannot be placed inside or outside** the pentagon without a crossover.

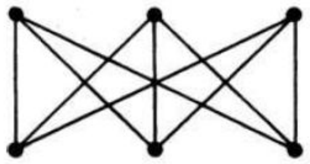
Thus the graph cannot be embedded in a plane.



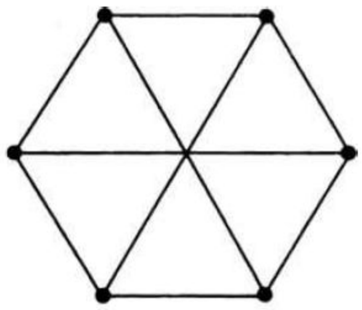
KURATOWSKI’S TWO GRAPHS

The complete graph of five vertices is the first of the two graphs of Kuratowski.

The second graph of Kuratowski is a regular connected graph with six vertices and nine edges, shown in its two common geometric representations here. This is also a nonplanar graph, as stated in **THEOREM 5-2**, which can be proved using the same arguments as **THEOREM 5-1**.



(a)



(b)

KURATOWSKI'S TWO GRAPHS

You may have noticed several properties common to the two graphs of Kuratowski:

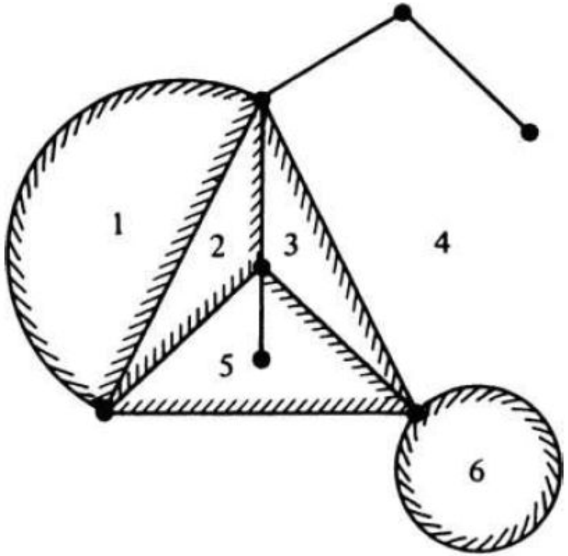
1. Both are regular graphs.
2. Both are nonplanar.
3. Removal of one edge or a vertex makes each a planar graph.
4. Kuratowski's first graph is the nonplanar graph with the **smallest number of vertices**, and Kuratowski's second graph is the nonplanar graph with the **smallest number of edges**. Thus both are the simplest nonplanar graphs.

In the literature, Kuratowski's first graph is usually denoted by K_5 and the second graph by $K_{3,3}$—letter K being for Kuratowski.

Regions or Faces

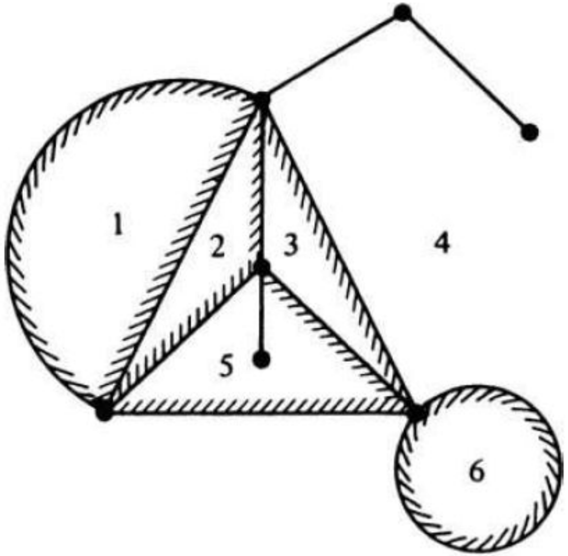
A plane representation of a graph divides the plane into **regions** (also called faces, or meshes).

A region is characterized by the set of edges (or the set of vertices) forming its boundary.



Regions or Faces

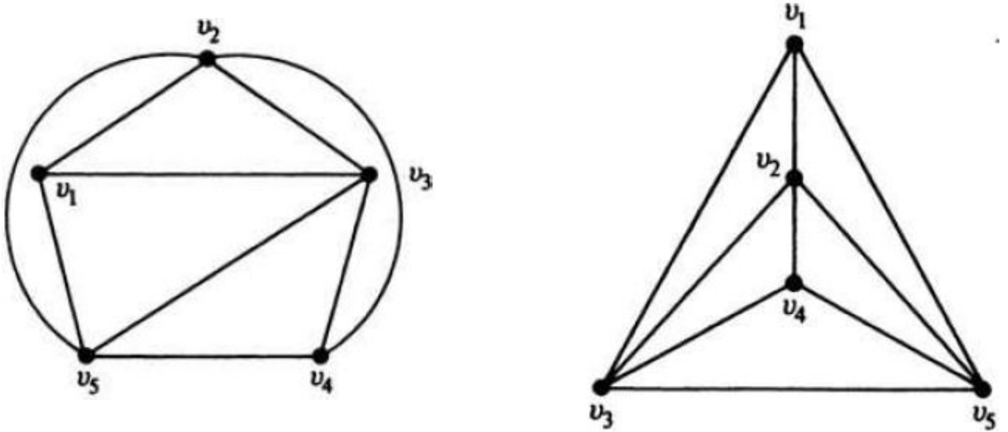
Infinite Region: The portion of the plane lying outside a graph embedded in a plane, such as region 4, is infinite in its extent. Such a region is called the infinite, unbounded, outer, or exterior region for that particular plane representation.



Regions or Faces

Like other regions, the infinite region is also characterized by a set of edges (or vertices). By changing the embedding of a given planar graph, we can change the infinite region.

For instance, following figure shows two different embeddings of the same graph, where the finite region defined by v_1, v_3, v_5 in the left graph becomes infinite region in the right graph.

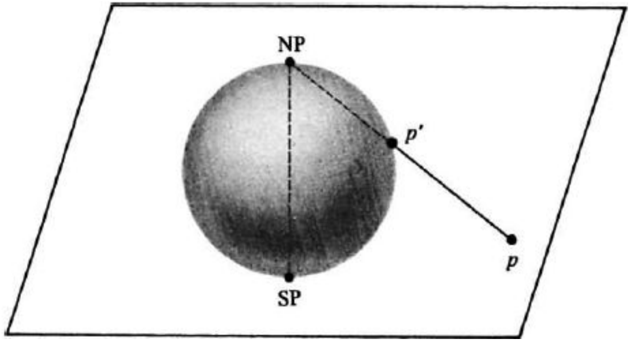


Embedding on a Sphere

THEOREM 5-4: ***"A graph can be embedded in the surface of a sphere if and only if it can be embedded in a plane."***

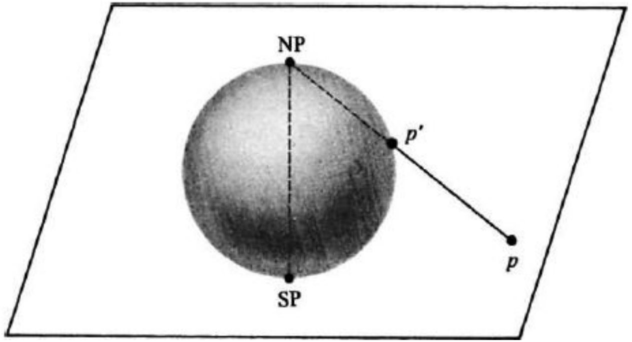
The fact that this theorem holds can be shown by stereographic projection of a sphere on a plane.

Put the sphere on the plane and call the point of contact SP (south pole). At point SP, draw a straight line perpendicular to the plane, and let the point where this line intersects the surface of the sphere be called NP (north pole).



Embedding on a Sphere

Now, corresponding to any point p on the plane, there exists a unique point p′ on the sphere and vice versa, where p′ is the point at which the straight line from point p to point NP intersects the surface of the sphere. Thus there is a one-to-one correspondence between the points of the sphere and the finite points on the plane, and points at infinity in the plane correspond to the point NP on the sphere.



Embedding on a Sphere

THEOREM 5-5: ***"A planar graph may be embedded in a plane such that any specified region (i.e., specified by the edges forming it) can be made the infinite region."***

A planar graph embedded in the surface of a sphere divides the surface into different regions. Each region on the sphere is finite, the infinite region on the plane having been mapped onto the region containing the point NP.

Now it is clear that by suitably rotating the sphere we can make any specified region map onto the infinite region on the plane.

Embedding on a Sphere

Thinking in terms of the regions on the sphere, we see that there is no real difference between the infinite region and the finite regions on the plane. Therefore, when we talk of the regions in a plane representation of a graph, we include the infinite region.

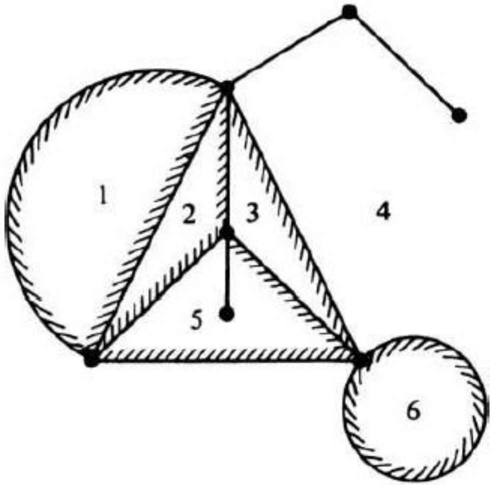
Also, since there is no essential difference between an embedding of a planar graph on a plane or on a sphere

Euler’s Formula

THEOREM 5-6: ***“A connected planar graph with n vertices and e edges has (e−n+2) regions.”***

It will suffice to prove the theorem for a simple graph, because adding a self-loop or a parallel edge simply adds one region to the graph and simultaneously increases the value of e by one.

We can also disregard (i.e., remove) all edges that do not form boundaries of any region. Three such edges are shown in the graph here. Addition (or removal) of any such edge increases (or decreases) e by one and also increases (or decreases) n by one, keeping the RHS of the equation unaltered.



Euler’s Formula

According to **Theorem 5-3**, any simple planar graph can have a plane representation such that each edge is a straight line, thus any planar graph can be drawn such that each region is a polygon (a polygonal net).

Let the polygonal net representing the given graph consist of f regions or faces, and let k_p be the number of p-sided regions. Since each edge is on the boundary of exactly two regions, so,

$$3 \cdot k_3 + 4 \cdot k_4 + 5 \cdot k_5 + \cdots + r \cdot k_r = 2 \cdot e,$$

where k_r is the number of polygons, with maximum edges. Also,

$$k_3 + k_4 + k_5 + \cdots + k_r = f.$$

Euler’s Formula

The sum of all angles subtended at each vertex in the polygonal net is $2\pi n$.

Recalling that the sum of all interior angles of a p-sided polygon is π(p − 2), and the sum of the exterior angles is π(p + 2), let us compute the expression as the grand sum of all interior angles of f − 1 finite regions plus the sum of the exterior angles of the polygon defining the infinite region. This sum is

$$\begin{aligned} &\pi(3-2)\cdot k_3 + \pi(4-2)\cdot k_4 + \cdots + \pi(r-2)\cdot k_r + 4\pi \\ &= \pi(2e-2f) + 4\pi. \end{aligned}$$

And this is equal to the sum of all angles subtended at each vertex, so,

2π(e − f) + 4π = 2πn,

or e − f + 2 = n.

Therefore, the number of regions is f = e − n + 2

Euler’s Formula

If the previous proof looks too geometric, then you can alternatively prove it using the logic of induction.

Apply the theorem for trivial cases, like a graph of 1 vertex or 1 edge (2 vertices). For both cases, f = 1, which follows the theorem as e-n+2 also gives 1.

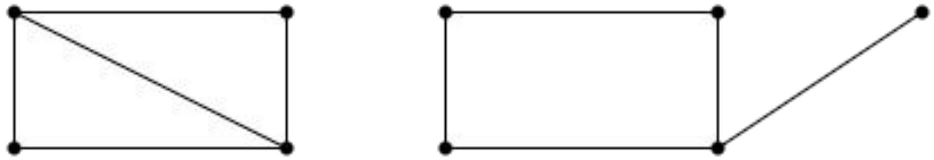
Now assume that the theorem will similarly hold for all graphs of upto k vertices/edges, that is $f_k = e_k - n_k + 2$, or equivalently, $e_k - n_k + f_k = 2$

And then we need to prove that the theorem holds for k+1 edges (or k+1 vertices).

Euler’s Formula

As an example, we will perform induction on the edges.

Now, for the k+1 edges case, there are two possibilities which are shown below:



For the left case, the new edge is added connecting two existing vertices, which divides the polygon in two regions but does not add any vertices, so

$f_{k+1} = f_k + 1$ and $n_{k+1} = n_k$

For the right case, the new edge is added connecting an existing vertex with a new vertex, which creates no new region but requires adding a new vertex, so

$f_{k+1} = f_k$ and $n_{k+1} = n_k + 1$

Euler's Formula

Substituting the values in the equation, for the left case,

LHS = $f_{k+1} = f_k + 1 = (e_k - n_k + 2) +1 = e_k - n_k + 3$

RHS = $e_{k+1} - n_{k+1} + 2 = (e_k + 1) - n_k + 2 = e_k - n_k + 3$

Similarly, substituting for the right case,

LHS = $f_{k+1} = f_k = e_k - n_k + 2$

RHS = $e_{k+1} - n_{k+1} + 2 = (e_k + 1) - (n_k + 1) + 2 = e_k - n_k + 2$

In both cases, LHS = RHS.

Corollary from Euler’s Formula

In any simple, connected planar graph with f regions, n vertices, and e edges (e>2), the following inequalities must hold :

$$e \geq \frac{3}{2}f,$$

$$e \leq 3n - 6.$$

Proof for these two inequalities is rather simple. Since each region is bounded by **at least three edges** and each edge belongs to **exactly two regions**, so,

$$2e \geq 3f$$

$$e \geq \frac{3}{2}f.$$

Corollary from Euler's Formula

Substituting for f from Euler's formula,

$$e \geq \frac{3}{2}(e - n + 2)$$

$$e \leq 3n - 6.$$

These inequalities are often useful in finding out if a graph is nonplanar. For example, in the case of K_5, the complete graph of five vertices,

$n = 5$, $e = 10$, $3n - 6 = 9 < e$.

Thus the graph violates this inequality and hence it is not planar.

Corollary from Euler's Formula

It should be noted that the previous inequality we just proved is only a necessary, but not a sufficient, condition for the planarity of a graph. In other words, although every simple planar graph must satisfy this inequality, the mere satisfaction of this inequality does not guarantee the planarity of a graph.

For example, Kuratowski's second graph, $K_{3.3}$, satisfies this inequality as

$e = 9$

$3n - 6 = 3 \cdot 6 - 6 = 12.$

So, $e < 3n - 6$, yet the graph is nonplanar.

Plane Representation and Connectivity

In a disconnected graph the embedding of each component can be considered independently. Therefore, it is clear that a disconnected graph is planar if and only if each of its components is planar.

Similarly, in a separable (or 1-connected) graph the embedding of each block (i.e., maximal nonseparable subgraph) can be considered independently. Hence a separable graph is planar if and only if each of its blocks is planar.

Therefore, in questions of embedding or planarity, one need consider only nonseparable graphs.

DETECTION OF PLANARITY

How to tell if a given graph is planar or non-planar?

There may be many possible geometric representations of a graph. Looking at all possible such representations to detect planarity is not an efficient way of answering this.

Here we will discuss an efficient way of detecting planarity of a graph, the first step of which is simplifying the graph by **Elementary Reduction**.

DETECTION OF PLANARITY

Step 1: Since a disconnected graph is planar if and only if each of its components is planar, we need consider only one component at a time. Also, a separable graph is planar if and only if each of its blocks is planar. So, if we have $G = \{G_1, G_2, \dots, G_k\}$, then we have to test each G_i for planarity.

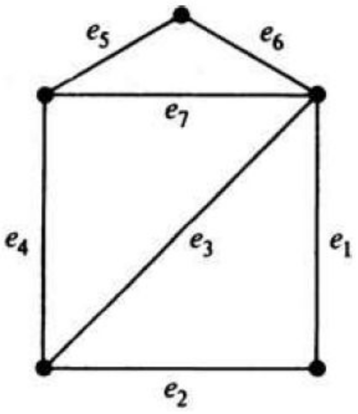
Step 2: Since addition or removal of self-loops does not affect planarity, remove all self-loops.

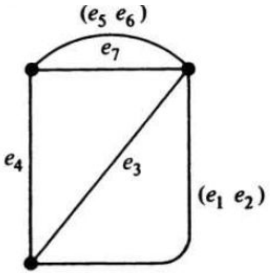
Step 3: Since parallel edges also do not affect planarity, eliminate edges in parallel by removing all but one edge between every pair of vertices.

Step 4: Elimination of a vertex of degree two by merging two edges in series does not affect planarity. Therefore, eliminate all edges in series.

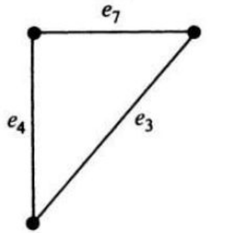
DETECTION OF PLANARITY

Repeated application of steps 3 and 4 will usually reduce a graph drastically. For example, the graph below is reduced by following the Elementary Reduction steps.

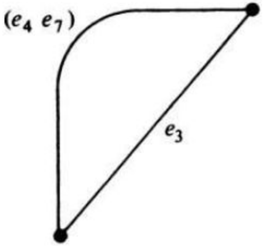




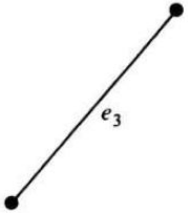
(a) Series Reduced



(b) Parallel Reduced



(c) Series Reduced



(d) Parallel Reduced

DETECTION OF PLANARITY

Let the nonseparable connected graph G_i be reduced to a new graph H_i after the repeated application of steps 3 and 4. What are the possible status of this H_i?

Graph H_i is

1. A single edge, or
2. A complete graph of four vertices, or
3. A nonseparable, simple graph with $n \geq 5$ and $e \geq 7$.

All H_i falling in categories 1 or 2 are planar and need not be checked further.

DETECTION OF PLANARITY

For category 3, we can first check to see if $e \leq 3n - 6$.

If this inequality is not satisfied, the graph H_i is nonplanar.

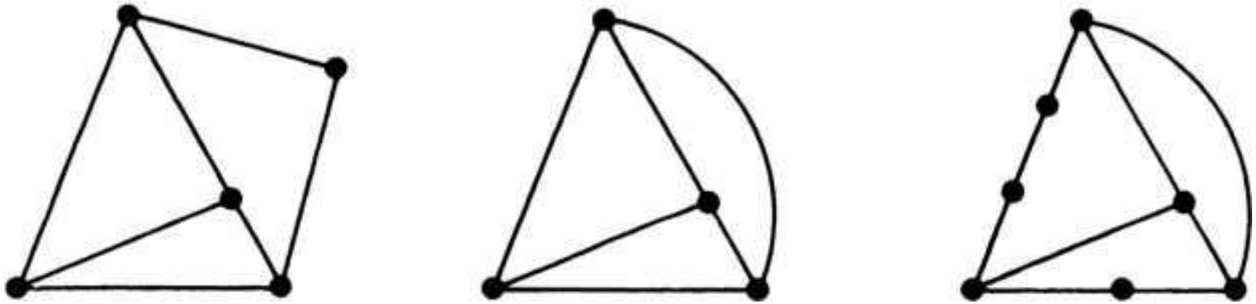
If the inequality is satisfied, we have to test the graph further.

The necessary and sufficient condition for a graph G to be planar is that G does not contain either of Kuratowski's two graphs or any graph **homeomorphic** to either of them.

Homeomorphic Graphs

Two graphs are said to be homeomorphic if one graph can be obtained from the other by the creation of edges in series (i.e., by insertion of vertices of degree two) or by the merger of edges in series.

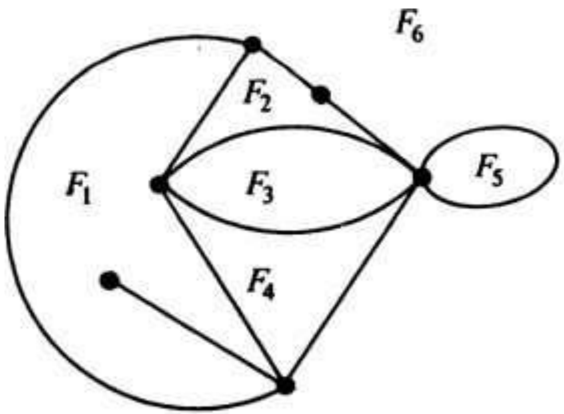
The three graphs shown here are homeomorphic.

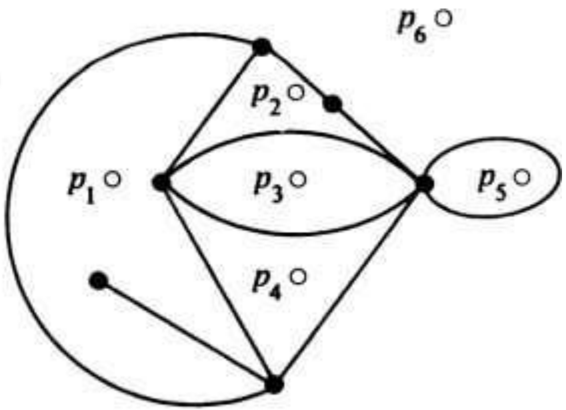


GEOMETRIC DUAL

Consider a graph with six regions or faces F_1, F_2, …, F_6.

Let us place six points p_1, p_2, … , p_6, one in each of the regions, as shown below:





GEOMETRIC DUAL

Next let us join these six points according to the following procedure:

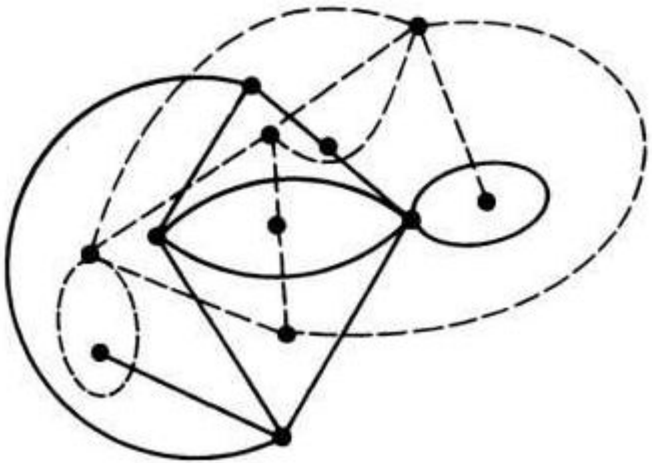
If two regions F_i and F, are adjacent (i.e., have a common edge), draw a line joining points p_i and p_j that intersects the common edge between F_i and F_j exactly once.

If there is more than one edge common between F_i and F_j, draw one line between points p_i and p_j for each of the common edges.

For an edge e lying entirely in one region, say F_k, draw a self-loop at point p_k intersecting e exactly once.

GEOMETRIC DUAL

By this procedure we obtain a new graph G* (shown in dashed lines), which is called the **Geometric Dual** of G.



Observations regarding GEOMETRIC DUAL

Clearly, there is a one-to-one correspondence between the edges of graph G and its dual G*, one edge of G* intersecting one edge of G.

Some simple observations that can be made about the relationship between a planar graph G and its dual G* are:

1. An edge forming a self-loop in G yields a pendant edge in G*.

2. A pendant edge in G yields a self-loop in G*.

3. Edges that are in series in G produce parallel edges in G*.

4. Parallel edges in G produce edges in series in G*.

Observations regarding GEOMETRIC DUAL

5. Remarks 1-4 are the result of the general observation that the number of edges constituting the boundary of a region Fi in G is equal to the degree of the corresponding vertex pi in G*, and vice versa.

6. Graph G* is also embedded in the plane and is therefore planar.

7. Considering the process of drawing a dual G* from G, it is evident that G is a dual of G*. Therefore, instead of calling G* a dual of G, we usually say that G and G* are dual graphs.

Observations regarding GEOMETRIC DUAL

8. If n, e, f, r, and μ denote as usual the numbers of vertices, edges, regions, rank, and nullity of a connected planar graph G, and if n^*, e^*, f^*, r^*, and μ^* are the corresponding numbers in dual graph G^*, then

$n^* = f$,

$e^* = e$,

$f^* = n$.

Using the above relationship, one can immediately get

$r^* = \mu$,

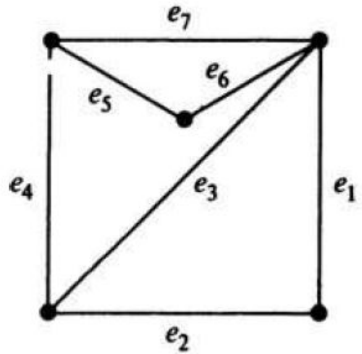
$\mu^* = r$.

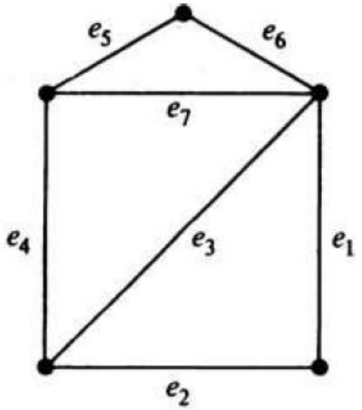
Uniqueness of Dual Graphs

Are all duals of a given graph isomorphic?

From the method of constructing a dual, it is reasonable to expect that a planar graph G will have a unique dual if and only if it has a unique plane representation or unique embedding on a sphere.

For example, construct the duals of the following two isomorphic graphs:



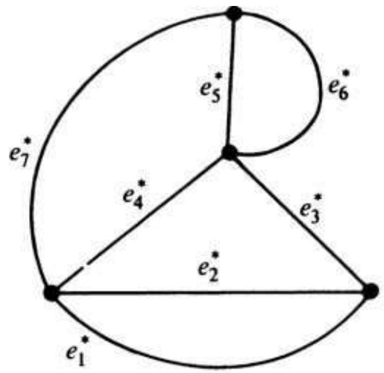


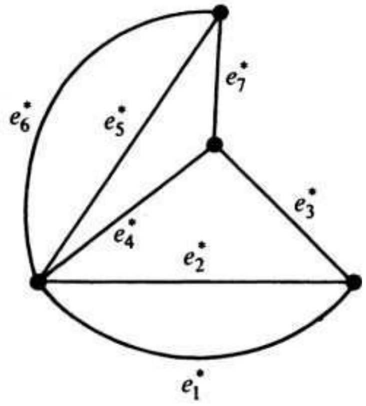
Uniqueness of Dual Graphs

The following two graphs are the duals of the previous two embeddings of the same graph. Note that these two dual graphs are non-isomorphic.

However, they are 2-isomorphic.

THEOREM 5-10: "*All duals of a planar graph G are 2-isomorphic; and every graph 2-isomorphic to a dual of G is also a dual of G.*"





COMBINATORIAL DUAL

So far we have defined and discussed duality of planar graphs in a purely geometric sense. The following theorem provides us with an equivalent definition of duality independent of geometric notions.

THEOREM 5-11: ***“A necessary and sufficient condition for two planar graphs G_1 and G_2 to be duals of each other is as follows: There is a one-to-one correspondence between the edges in G1 and the edges in G2 such that a set of edges in G_1 forms a circuit if and only if the corresponding set in G2 forms a cut-set.”***

Dual of a Subgraph

Let G be a planar graph and G^* be its dual. Let a be an edge in G, and the corresponding edge in G^* be a^*.

Suppose that we delete edge a from G and then try to find the dual of $G - a$. If edge a was on the boundary of two regions, removal of a would merge these two regions into one. Thus the dual $(G - a)^*$ can be obtained from G^* by deleting the corresponding edge a^* and then fusing the two end vertices of a^* in $G^* - a^*$.

On the other hand, if edge a is not on the boundary, a^* forms a self-loop. In that case $G^* - a^*$ is the same as $(G - a)^*$.

Thus if a graph G has a dual G^*, the dual of any subgraph of G can be obtained by successive application of this procedure.

Dual of a Homeomorphic Graph

Let G be a planar graph and G* be its dual. Let a be an edge in G, and the corresponding edge in G* be a*.

Suppose that we create an additional vertex in G by introducing a vertex of degree two in edge a (i.e., a now becomes two edges in series). How will this addition affect the dual?

It will simply add an edge parallel to a* in G*. Likewise, the reverse process of merging two edges in series will simply eliminate one of the corresponding parallel edges in G*.

Thus if a graph G has a dual G*, the dual of any graph homeomorphic to G can be obtained from G* by the above procedure.

THEOREM 5-12

"A graph has a dual if and only if it is planar."

We need prove just the "only if" part. That is, we have only to prove that a nonplanar graph does not have a dual.

Let G be a nonplanar graph. Then according to Kuratowski's theorem, G contains K_5 or $K_{3,3}$ or a graph homeomorphic to either of these.

Thus if we can show that neither K_5 nor $K_{3,3}$ has a dual, we have proved the theorem. We shall prove this by contradiction.

THEOREM 5-12

Suppose that $K_{3,3}$ has a dual D. Observe that the cut-sets in $K_{3,3}$ correspond to circuits in D and vice versa (by THEOREM 5-10).

Since $K_{3,3}$ has no cut-set consisting of two edges, D has no circuit consisting of two edges. That is, D contains no pair of parallel edges.

Since every circuit in $K_{3,3}$ is of length four or six, D has no cut-set with less than four edges. Therefore, the degree of every vertex in D is at least four.

As D has no parallel edges and the degree of every vertex is at least four, D must have at least five vertices each of degree four or more.

That is, D must have at least $(5 \times 4)/2 = 10$ edges. This is a contradiction, because $K_{3,3}$ has nine edges and so must its dual. Thus $K_{3,3}$ cannot have a dual.

Using similar arguments, you can also prove that K_5 cannot have a dual.

THICKNESS AND CROSSINGS

If a graph is non-planar, then the next logical question is what is the minimum number of planes necessary for embedding G?

The least number of planar subgraphs whose union is the given graph G is called the **thickness** of G. In a printed-circuit board, for instance, the number of insulation layers necessary is the thickness of the corresponding graph.

By definition, the thickness of a planar graph is one. The thicknesses of Kuratowski's 1st and 2nd graphs are clearly two. The thickness of the complete graph of eight vertices is two, while the thickness of the complete graph of nine vertices is three.

The thickness of an arbitrary graph is in general, difficult to determine.

THICKNESS AND CROSSINGS

Another question one might ask about a nonplanar graph is: **What is the fewest number of crossings (or intersections) necessary in order to “draw” the graph in a plane?**

The crossing number of a planar graph is, by definition, zero.

For both Kuratowski’s graphs, it is one.

The crossing numbers of only a few graphs have been determined. No formula exists to give the crossing number of an arbitrary graph.